ENCHANTER

THE CIRCLE OF ENCHANTERS

A maximum-score route to the triumphant ending

SCORE	COMMANDS	DEATHS	INSTRUCTIONS
400 / 400	233	0	503,562

Verified story: Z-machine V3, Release 29, Serial 860820. The route defeats Krill, reaches the Circle ending at 400/400, and records no deaths. It was replayed from a fresh game with deterministic seed 1.

How to use this guide
Follow the numbered stages in order. Commands are shown in Courier. Directions assume the route leaves you in the stated room. The turtle-fetch, adventurer, and map sequences are included as tested. Eat and drink when the route says to, and rest in the Bedroom and Map Room.

Route checkpoints
The replay consumes 233 command inputs and 503,562 VM instructions. It reaches 400/400, eliminates Krill with Guncho, and halts after the Circle ending. The supplied local regression route records the exact command sequence and opcode trace.

Story fingerprint: SHA-256 BC9B99BE980A1AA67C1AEF89D4E5878F1F03EDAB62FE43AE23B2F7E2C41B1DC8

Opcode sequence numbers are 1-based. Program counters are hexadecimal byte addresses. The opcode inventory records every unique instruction observed in the complete replay.

01. Supplies and the castle gate

Collect food and water outside the castle. The crone supplies Rezrov; copy it into the spell book and use it to open the western gate.

```
ne; enter shack; take jug; open oven; take loaf; south
ne; se; ne; fill jug
sw; se; sw; sw; s; read scroll; gnusto rezrov
ne; ne; e; e; learn rezrov; rezrov gate; e
```

02. The repaired Zifmia scroll

Open the egg in the Jewel Room, then use the crumpled forest scroll to repair the damaged Zifmia scroll. Copy Zifmia into the book for the later summoning.

```
frotz spell book; n; n; u; take egg; learn rezrov; rezrov egg; take shredded scroll
d; e five times; learn rezrov; rezrov gate; n
take crumpled scroll; read crumpled scroll; krebf shredded scroll
read faded scroll; gnusto zifmia
```

03. Learn Gondar

Find the rat hole in the Library, retrieve the frayed scroll, and copy Gondar into the spell book. Leave north if the hairy shapes approach.

```
s; s; examine rat tracks; reach into hole; n
read frayed scroll; gnusto gondar
```

04. Vaxum and Ozmoo

Open the bedpost compartment for Vaxum. Use the spell book as a light, then reveal and read the black scroll behind the lighted portrait.

```
w five times; s four times; u; examine bedpost; learn rezrov; rezrov bedpost
take gold leaf scroll; read gold leaf scroll; gnusto vaxum
d; e; drop spell book; e; remove lighted portrait; take black scroll; read black scroll
w; take spell book; drink water
```

05. Exex, the turtle, and Kulcad

Find Exex behind the loose cell block. Convince the turtle to fetch the brittle Kulcad scroll. Rest in the Bedroom, then return to the Engine Room and collect it before leaving.

```
d; open cell door; n; examine writing; remove block; e; take stained scroll; read stained scroll; eat loaf
w; s; u; e; e; s; se; learn nitfol; nitfol turtle; turtle, follow me
nw; n; e; u; exex turtle; turtle, go se and get scroll; yell at turtle
d; w four times; u; lie down; wait; get up
d; e four times; u; take brittle scroll; read brittle scroll
d; w four times; u; d; n; n; e; e; drink water
```

06. Survive the sacrifice and get Melbor

Protect yourself with Ozmoo, drop your inventory to be captured, then recover it after the protected sacrifice. The dagger opens the jewelled box.

```
ozmoo me; drop all; e; wait; d; w; get all
w; s; cut rope; open box; take vellum scroll; read vellum scroll; drop dagger
```

07. Recruit the adventurer; get the map

Summon the adventurer from the mirror after he appears, make him friendly with Vaxum, and show him the egg. At the Map Room, drop the egg and take the map while the adventurer is present; take the pencil too.

```
n; w; n; n; e; learn vaxum; e; w; learn zifmia; zifmia adventurer
vaxum adventurer; show egg to adventurer; e five times; point at door; n
drop egg; take map; take pencil; eat loaf; drink water; lie down; wait; get up
```

08. The safe route to Guncho

Use the map and pencil in the Dungeon to seal the Terror away from the powerful scroll. After editing the map, drop both tools before entering the maze. Drop the remaining food and water after leaving the maze.

```
d; w six times; s four times; e; d
connect b and k; connect p and f; wait twice; disconnect j and v; disconnect b and k
drop map; drop pencil; d; s; e; ne; se; sw; take powerful scroll; read powerful scroll
ne; nw; sw; w; n; u; learn gondar; learn vaxum; drop jug; drop loaf
```

09. Defeat Krill

Use Melbor against the temple creatures, then cast Kulcad on the false stair. The falling sequence reveals Izyuk; fly east to Krill and use the memorized combat spells and Guncho scroll.

```
u; e; e; n; melbor me; n; n; e; e
learn gondar; learn vaxum; kulcad stairs; read ornate scroll; izyuk me; e
gondar dragon; vaxum shape; guncho krill; quit
```

Completion check: The ending should say that Krill is gone and you have joined the Circle. The final score is 400/400; there are no deaths.

Executed opcode inventory

58 unique opcodes were observed across 503,562 instructions in the fresh-game route. Counts include all calls and story output. Opcode values are hexadecimal; the form disambiguates encodings. PCs are byte addresses.

Form	Opcode	Mnemonic	Count	First seq.	First PC
Variable	0x02	storeb	42,508	1	0x00550B
Variable	0x00	call	22,605	2	0x005510
Long	0x14	add	30,858	4	0x005477
Long	0x01	je	36,529	6	0x00547F
Long	0x15	sub	4,922	7	0x005483
Short	0x00	jz	69,505	8	0x005487
Variable	0x01	storew	17,802	10	0x005492
Short	0x0B	ret	2,075	11	0x005497
Long	0x0F	loadw	26,406	20	0x005499
Short	0x0C	jump	16,534	23	0x0054A7
Long	0x0D	store	18,542	246	0x005579
Long	0x0A	test_attr	33,917	250	0x005585
Short	0x02	print	648	251	0x00558A
Short	0x0B	new_line	613	252	0x005875
Variable	0x09	and	1	257	0x007FE4
Variable	0x06	print_num	4	258	0x007FEA
Long	0x05	inc_chk	13,143	260	0x007FFC
Long	0x10	loadb	47,684	261	0x008003
Variable	0x05	print_char	6	262	0x008007
Short	0x00	rtrue	3,996	287	0x00800E
Long	0x0E	insert_obj	292	289	0x00587D
Variable	0x08	push	2,720	294	0x0079A4
Variable	0x09	pull	268	296	0x0079AD
Long	0x0B	set_attr	361	299	0x0079D7
Long	0x06	jin	2,047	301	0x0079DD
Short	0x0A	print_obj	233	302	0x0079E1
Short	0x03	get_parent	3,527	305	0x0079EA
Long	0x11	get_prop	2,248	308	0x007A05
Short	0x0D	print_paddr	162	317	0x007A89
Long	0x03	jg	4,570	322	0x007AAC
Long	0x02	jl	2,692	324	0x007AEC
Short	0x02	get_child	5,578	333	0x007B1C
Variable	0x01	je	7,382	346	0x007C75
Short	0x01	get_sibling	9,162	351	0x007CD4
Short	0x08	ret_popped	2,112	364	0x007B38
Variable	0x04	sread	233	466	0x005EBB

Form	Opcode	Mnemonic	Count	First seq.	First PC
Long	0x16	mul	14,850	483	0x00643C
Long	0x04	dec_chk	29,054	835	0x006480
Long	0x07	test	2,663	1,223	0x0064F5
Long	0x09	and	743	1,225	0x0064FD
Long	0x12	get_prop_addr	10,529	1,292	0x0086C9
Short	0x04	get_prop_len	2,559	1,294	0x0086D1
Short	0x05	inc	971	1,517	0x0054D2
Variable	0x0D	store	317	2,581	0x0071CC
Long	0x0C	clear_attr	169	2,585	0x0071E3
Long	0x17	div	1,821	2,618	0x007853
Short	0x01	rfalse	7,223	2,628	0x007756
Short	0x03	print_ret	100	4,944	0x0088A2
Variable	0x07	random	140	8,477	0x017537
Short	0x06	dec	76	8,523	0x00541F
Short	0x09	remove_obj	26	14,912	0x00AE00
Variable	0x03	put_prop	63	14,934	0x00AEC4
Short	0x0E	load	2	156,060	0x008711
Long	0x18	mod	4	198,381	0x01602B
Short	0x07	print_addr	1	276,610	0x006F13
Long	0x13	get_next_prop	364	334,926	0x010FCD
Short	0x0C	show_status	1	503,533	0x00D688
Short	0x0A	quit	1	503,562	0x007EEC